

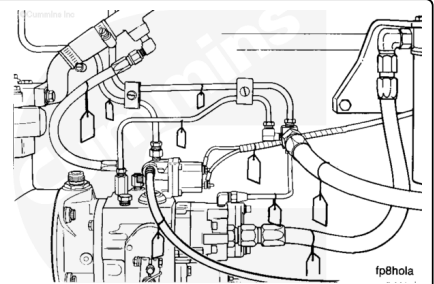
## Trainings ▾

### Preparatory Steps

#### Industrial and Power Generation

##### **⚠ WARNING ⚠**

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



##### **⚠ WARNING ⚠**

Coolant is toxic. Keep away from small children and pets. If not reused, dispose of in accordance with local environmental regulations.

##### **⚠ WARNING ⚠**

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

##### **⚠ WARNING ⚠**

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

##### **⚠ WARNING ⚠**

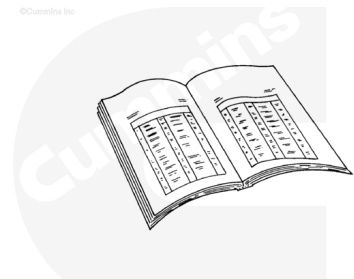
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.

- Attach a tag to all hoses, lines, linkages, and electrical connections, as they are removed, to identify location and aid the installation process.
- Disconnect the battery cables.
- Drain the engine coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html) (Cooling System) in Section 8.
- Drain the lubricating oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html) (Lubricating Oil System) in Section 7.

## QSK60 Marine Applications

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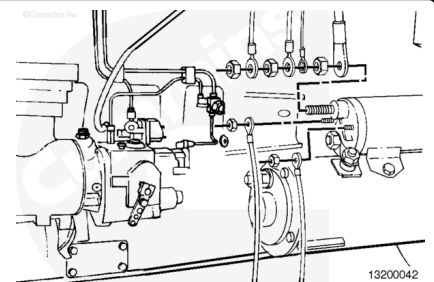
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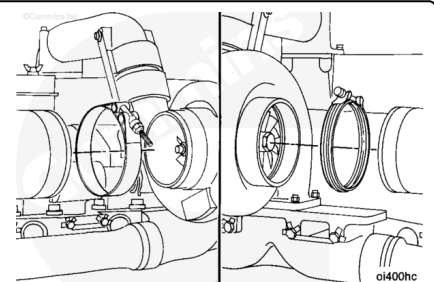
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- Drain the engine coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>) (Cooling System) in Section 8.
- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>) (Lubricating Oil System) in Section 7.
- Remove aftercooler heat shields. Refer to Procedure 010-129 (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>) (Aftercooler Heat Shield) in Section 10.
- Remove the turbocharger heat shields. Refer to Procedure 010-076 (</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>) (Turbocharger Heat Shield) in Section 10.
- Remove the turbocharger inlet connections. Refer to Procedure 010-130 (</qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html>) (Turbocharger Inlet Connections) in Section 10.

## Remove

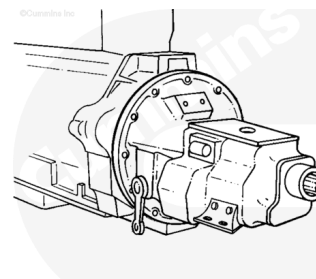
Disconnect the starter cable, engine ground straps, cab or chassis to engine hoses, tubing, electrical wires, and hydraulic lines.



Disconnect the intake and exhaust system pipes.

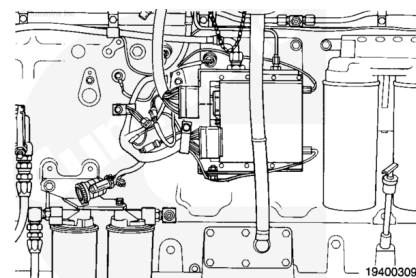


Disconnect the drive units from the flywheel housing and flywheel.



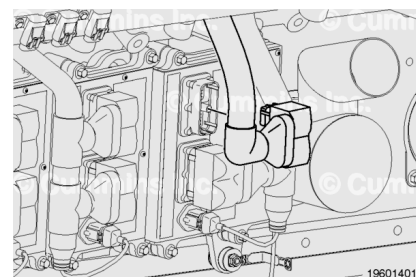
Disconnect the OEM wiring harness from the ECM.

- For QSK45 engines, refer to the Troubleshooting and Repair Manual, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.



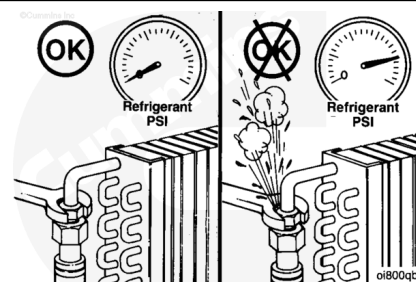
### with Electronically Actuated Injector

- For QSK60 engines, refer to the Troubleshooting and Repair Manual, QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533.



### **⚠ WARNING ⚠**

If a liquid refrigerant system (air conditioning) is used, wear eye and face protection, and wrap a cloth around the fittings before removing. Liquid refrigerant can cause serious eye and skin injuries.

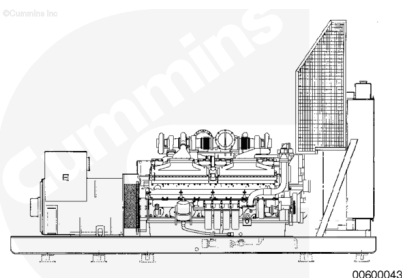


### **⚠ WARNING ⚠**

To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere. Federal law requires capturing and recycling the refrigerant.

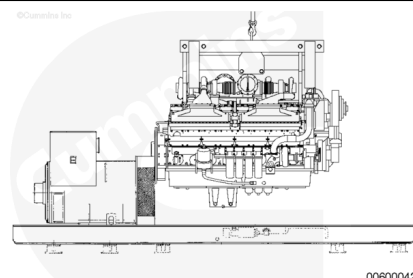
Remove air conditioning lines if needed.

Remove all chassis components necessary to remove the engine from the equipment.



### ⚠ WARNING ⚠

**Inspect the engine lifting brackets for damage or cracks. Do not attempt to lift the engine if any cracks or damage are visible. Using damaged lifting equipment can cause serious personal injury and/or property damage.**



### ⚠ WARNING ⚠

**The QSK45 and QSK60 series engines can weigh as much as 9980 kg [22,000 lb]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.**

The engine lifting fixtures, Part Number 3163264, are designed to lift the engine **only**. Any additional components attached (including to the drive end, non-drive end and/or chassis) **must** be removed.

For engine weights, Refer to Procedure 018-015  
 (/qs3/pubsys2/xml/en/procedures/56/56-018-015-tr.html)  
 (Specifications, General Engine) in Section V.

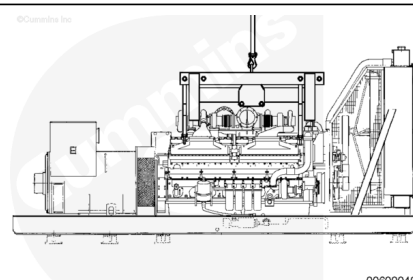
Inspect the engine lifting brackets for cracks.

The engine lifting equipment is designed to lift the engine safely.

The lifting fixture, Part Number 3163264, is designed to lift a maximum of 11,340 kg [25,000 lb].

Use a properly rated hoist and lifting fixture to support the weight of the engine. Remove the front and rear engine support capscrews.

Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to remove the engine.



The lifting force **must** be in a straight vertical line with the lifting brackets in both directions.

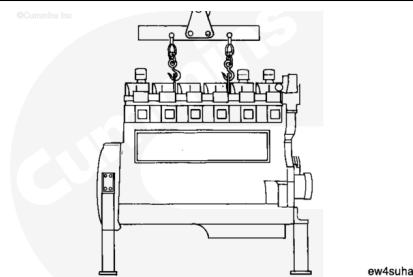
**⚠ WARNING ⚠**

**The oil sump or cover plate will not support the weight of the engine. The engine will fall and personal injury can result.**

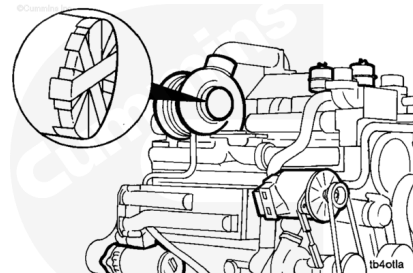
**⚠ WARNING ⚠**

**Use a stand or skid that contacts the engine mounts. The stand must hold the weight of the engine and provide permanent support to prevent the engine from falling.**

Place the engine on the stand.



Cover all engine openings to prevent dirt and debris from entering the engine.



Remove all remaining accessories and brackets that will be used on the replacement engine.

